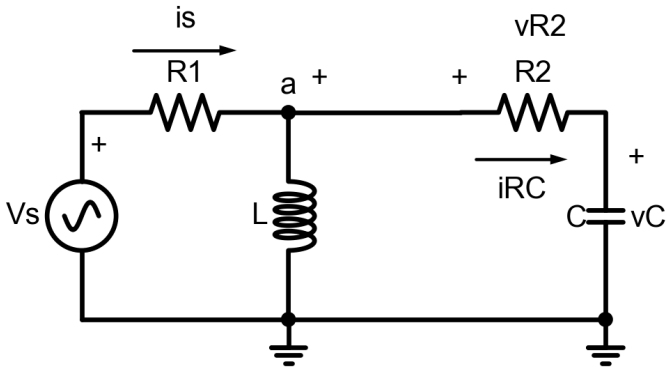
电电 1 期末考试样题

第 1 题

如下图所示电路，全部相量均采用 RMS 值。参数为：

$$v_s(t) = 10\sqrt{2}\cos(1000t)\,\mathrm{V}$$
$$R_1 = R_2 = 1\,\mathrm{k\Omega}, \qquad L = 1\,\mathrm{H}, \qquad C = 1\,\mu\mathrm{F}.$$



1. 写出 L、C、串联 $R_2 - C$ 支路以及节点 a 对地等效阻抗在 $\omega = 1000\,\mathrm{rad/s}$ 下的复阻抗。
2. 求电路总阻抗 Z_{eq}、电源电流相量 $\tilde{I}_s$ 和节点电压相量 $\tilde{V}_a$。
3. 只判断串联 $R_2 - C$ 支路中 v_{R_2} 与 v_C 的超前/滞后关系，并说明相位差。
4. 写出电源电流 $i_s(t)$ 的时域表达式。

For the circuit shown in the figure, all phasors use RMS values.

The values are

$$v_s(t) = 10\sqrt{2}\cos(1000t)\,\mathrm{V}$$
$$R_1 = R_2 = 1\,\mathrm{k\Omega}, \qquad L = 1\,\mathrm{H}, \qquad C = 1\,\mu\mathrm{F}.$$

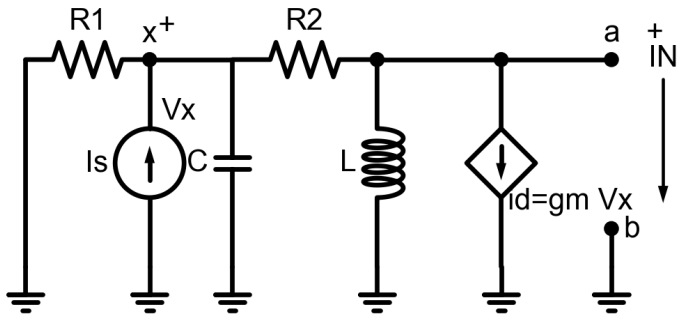
1. At $\omega = 1000\,\mathrm{rad/s}$, find the complex impedances of L, C, the series $R_2 - C$ branch, and the equivalent impedance from node a to ground.
2. Find the total circuit impedance Z_{eq}, the source current phasor $\tilde{I}_s$, and the node-voltage phasor $\tilde{V}_a$.
3. Only determine whether v_{R_2} leads or lags v_C in the series $R_2 - C$ branch, and state the phase difference.
4. Write the time-domain expression for the source current $i_s(t)$.

第 2 题

如下图所示电路工作在 $\omega = 10^4\,\mathrm{rad/s}$。端口为 $a - b$，其中 b 接地；求等效时端口外部负载未接入。所有相量均采用 RMS 值。其中：

$$\tilde{I}_s = 15\angle 0^\circ\,\text{mA}, \quad R_1 = R_2 = 1\,\text{k}\Omega, \quad C = 0.10\,\mu\text{F}, \quad L = 50\,\text{mH}, \quad g_m = 3\,\text{mS}.$$

$$i_d = g_m \tilde{V}_x.$$



1. 写出开路条件下节点 x 与节点 a 的相量 KCL 方程。
2. 求端口开路电压 $\tilde{V}_{\text{Th}} = \tilde{V}_a$。
3. 将独立电流源置零，用测试源法求端口等效阻抗 Z_{Th}，并写出诺顿等效的 I_{N} 与 Z_{N}。诺顿电流 I_{N} 取端口短路电流从 a 流向 b 为正。

The circuit shown in the figure operates at $\omega = 10^4\,\text{rad/s}$. The port is $a - b$, where node b is grounded; the external load is not connected when finding the equivalent circuit. All phasors use RMS values.

The values are

$$\tilde{I}_s = 15\angle 0^\circ\,\text{mA}, \quad R_1 = R_2 = 1\,\text{k}\Omega, \quad C = 0.10\,\mu\text{F}, \quad L = 50\,\text{mH}, \quad g_m = 3\,\text{mS}.$$

The dependent-source current is

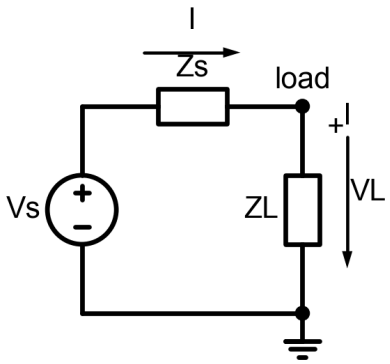
$$i_d = g_m \tilde{V}_x.$$

1. Write the phasor-domain KCL equations at nodes x and a under open-circuit conditions.
2. Find the open-circuit port voltage $\tilde{V}_{\text{Th}} = \tilde{V}_a$.
3. Set the independent current source to zero and use the test-source method to find the port equivalent impedance Z_{Th}. Then give the Norton equivalent I_{N} and Z_{N}. The positive reference direction of I_{N} is the short-circuit current from a to b.

第 3 题

在正弦稳态下，考虑如下图所示电路。

$$\tilde{V}_s = 60\angle 0^\circ\,\text{V}_{\text{rms}}, \qquad Z_s = 3 + j4\,\Omega.$$



1. 当 $Z_L = 2 - j4\,\Omega$ 时，求负载电流相量 $\tilde{I}$、负载端电压相量 $\tilde{V}_L$，以及负载复功率 $S_L = P_L + jQ_L$。
2. 若 $Z_L = R_L + jX_L$ 的实部和虚部均可调，写出使负载获得最大平均功率的阻抗匹配条件，并求 $Z_{L,\mathrm{opt}}$。
3. 在共轭匹配条件下，求负载可获得的最大平均功率 $P_{L,\max}$，并给出此时电流相量。
4. 若负载只能取纯电阻 R_L，求使 P_L 最大的 R_L 和此时最大平均功率，并与共轭匹配结果作定性比较。

In sinusoidal steady state, consider the circuit shown in the figure.

The values are

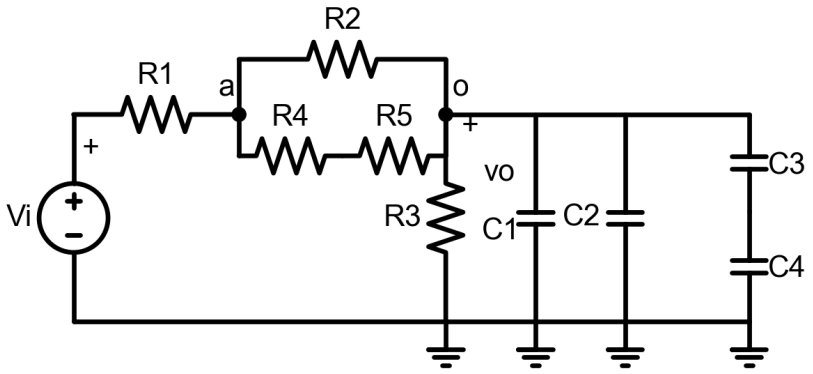
$$\tilde{V}_s = 60\angle 0^\circ\,\mathrm{V_{rms}}, \qquad Z_s = 3 + j4\,\Omega.$$

1. For $Z_L = 2 - j4\,\Omega$, find the load current phasor $\tilde{I}$, the load voltage phasor $\tilde{V}_L$, and the load complex power $S_L = P_L + jQ_L$.
2. If both the real and imaginary parts of $Z_L = R_L + jX_L$ are adjustable, state the impedance matching condition for maximum average power delivered to the load, and find $Z_{L,\mathrm{opt}}$.
3. Under the conjugate-matching condition, find the maximum average power available to the load, $P_{L,\max}$, and the corresponding current phasor.
4. If the load must be a pure resistance R_L, find the value of R_L that maximizes P_L, find the corresponding maximum average power, and qualitatively compare it with the conjugate-matching result.

第 4 题

如下图所示电路中，输出 v_o 取节点 o 对地电压。

$$R_1 = 3\,\mathrm{k\Omega}, \quad R_2 = R_3 = 6\,\mathrm{k\Omega}, \quad R_4 = R_5 = 3\,\mathrm{k\Omega},$$
$$C_1 = C_2 = C_3 = C_4 = 4\,\mathrm{nF}.$$



1. 将节点 a 到节点 o 的电阻网络化简，写出输入到输出节点的等效串联电阻 R_s；将输出端的电容网络化简，写出 C_{eq}。
2. 求输出端电容看到的戴维南电阻 R_{th}，写出传递函数 $H(s) = V_o(s)/V_i(s)$，并求直流增益、极点位置和截止角频率。
3. 定性说明幅频和相频 Bode 图的渐近趋势。
4. 若 $v_i(t) = 2u(t)\,\mathrm{V}$，且所有电容初始电压均为 0，写出输出电压 $v_o(t)$ 的时域表达式。

For the circuit shown in the figure, the output v_o is the voltage from node o to ground.

The values are

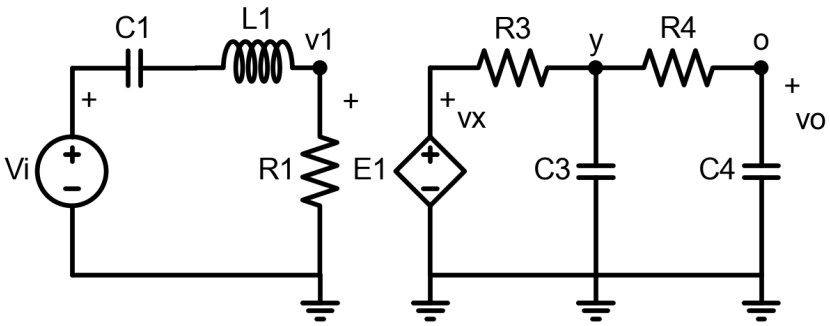
$$R_1 = 3\,\mathrm{k\Omega}, \quad R_2 = R_3 = 6\,\mathrm{k\Omega}, \quad R_4 = R_5 = 3\,\mathrm{k\Omega},$$
$$C_1 = C_2 = C_3 = C_4 = 4\,\mathrm{nF}.$$

1. Simplify the resistor network from node a to node o, and find the equivalent series resistance R_s from the input to the output node. Simplify the capacitor network at the output and find C_{eq}.
2. Find the Thevenin resistance R_{th} seen by the output capacitors. Write the transfer function $H(s) = V_o(s)/V_i(s)$, and find the DC gain, pole location, and cutoff angular frequency.
3. Qualitatively describe the asymptotic trends of the magnitude and phase Bode plots.
4. If $v_i(t) = 2u(t)\,\mathrm{V}$ and all capacitors have zero initial voltage, write the time-domain expression of $v_o(t)$.

第 5 题

如下图所示两级隔离网络中，各节点电压均以地为参考。各参数为：

$$R_1 = 100\,\Omega, \qquad L_1 = 100\,\mathrm{mH}, \qquad C_1 = 10\,\mu\mathrm{F}.$$
$$v_x = A_v v_1, \qquad A_v = 2.$$
$$R_3 = 5\,\mathrm{k\Omega}, \qquad R_4 = 10\,\mathrm{k\Omega}, \qquad C_3 = 10\,\mathrm{nF}, \qquad C_4 = 5\,\mathrm{nF}.$$



1. 分别写出第一级 $H_1(s) = V_1(s)/V_i(s)$、第二级 $H_2(s) = V_o(s)/V_x(s)$，并写出总传递函数 $H(s) = V_o(s)/V_i(s)$。
2. 求总传递函数的零点与极点位置。对复极点写出对应的 ω_0 与 Q。
3. 根据零极点说明该两级网络在低频、中频和高频处的增益变化趋势。
4. 定性描绘总传递函数的 Bode 幅频和相频图，并标出主要转折频率。

For the two-stage isolated network shown in the figure, all node voltages are referenced to ground. The values are

$$R_1 = 100\,\Omega, \qquad L_1 = 100\,\text{mH}, \qquad C_1 = 10\,\mu\text{F}.$$

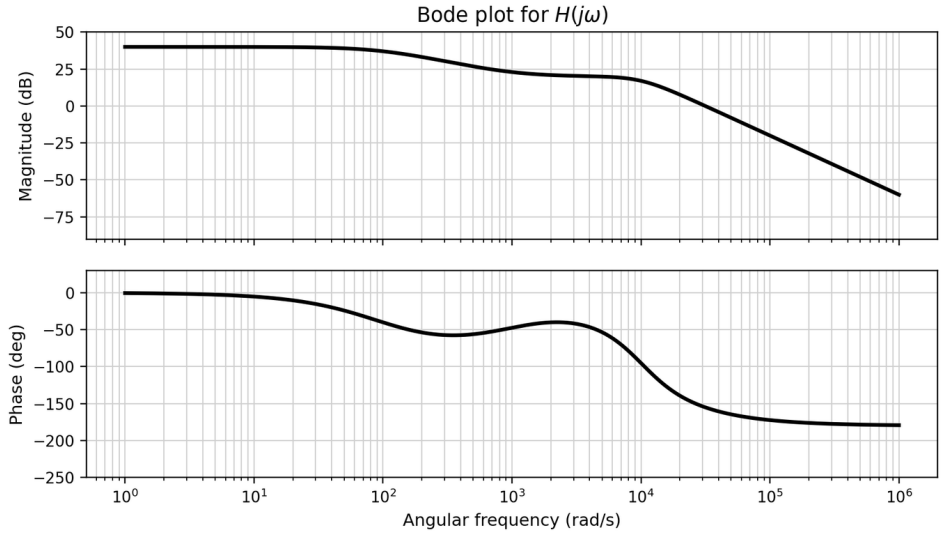
$$v_x = A_v v_1, \qquad A_v = 2.$$

$$R_3 = 5\,\text{k}\Omega, \qquad R_4 = 10\,\text{k}\Omega, \qquad C_3 = 10\,\text{nF}, \qquad C_4 = 5\,\text{nF}.$$

1. Find $H_1(s) = V_1(s)/V_i(s)$, $H_2(s) = V_o(s)/V_x(s)$, and the overall transfer function $H(s) = V_o(s)/V_i(s)$.
2. Find the zeros and poles of the overall transfer function. For the complex pole pair, give the corresponding ω_0 and Q.
3. Use the zeros and poles to explain the gain trend at low, middle, and high frequencies.
4. Sketch qualitatively the Bode magnitude and phase plots of the overall transfer function, and mark the main break frequencies.

第 6 题

如下图给出某系统频率响应 $H(j\omega)$ 的 Bode 图。所有零极点的自然频率为 10 的整数幂次。



1. 根据幅频渐近线，估计出 $H(s)$ 的传递函数形式，可保留为零极点因子形式。
2. 写出主要零点、极点的位置，假设 $Q = 1/\sqrt{2} \approx 0.707$。
3. 若该 $H(s)$ 是单位负反馈系统的环路响应，并且环路响应本身无右半平面极点，依据 Bode 图判断闭环系统是否稳定，并给出相位裕度的近似值。

The Bode plot in the figure gives the frequency response $H(j\omega)$ of a system. The natural frequencies of all zeros and poles are integer powers of 10.

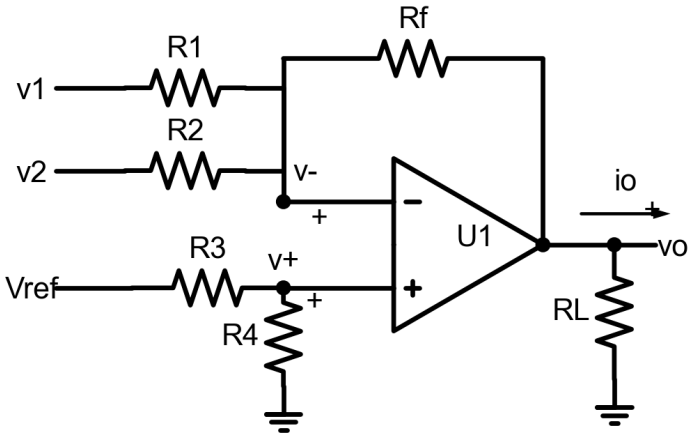
1. From the straight-line magnitude asymptotes, write an approximate transfer function $H(s)$. A zero-pole factor form is acceptable.
2. Give the main zeros and poles, assuming $Q = 1/\sqrt{2} \approx 0.707$.
3. If $H(s)$ is the loop response of a unity negative-feedback system, and the loop response itself has no right-half-plane poles, use the Bode plot to determine whether the closed-loop system is stable. Also give the approximate phase margin.

第 7 题

如下图所示理想运放电路，输出电流 i_o 取从运放输出端流出为正。

$$V_{\mathrm{ref}} = 4\,\mathrm{V}, \quad R_1 = 10\,\mathrm{k\Omega}, \quad R_2 = 15\,\mathrm{k\Omega}, \quad R_3 = R_4 = 10\,\mathrm{k\Omega}, \quad R_f = 30\,\mathrm{k\Omega}.$$

$$R_L = 1\,\mathrm{k\Omega}, \qquad V_{\mathrm{rail}} = \pm 10\,\mathrm{V}.$$



1. 在运放未饱的假设下，求 v_+，并推导 v_o 与 $v_1, v_2, V_{\mathrm{ref}}$ 的关系。
2. 当 $v_1 = 1\,\mathrm{V}$、$v_2 = 2\,\mathrm{V}$ 且未接入 R_L 时，求 v_o，并求此时运放的输出端电流。
3. 当 $v_1 = -1\,\mathrm{V}$、$v_2 = 0\,\mathrm{V}$ 且未接入 R_L 时，判断输出是否仍在线性区；若不在线性区，给出饱和方向和实际输出电压。

For the ideal op-amp circuit shown in the figure, the output current i_o is positive when flowing out of the op-amp output terminal.

The values are

$$V_{\mathrm{ref}} = 4\,\mathrm{V}, \quad R_1 = 10\,\mathrm{k\Omega}, \quad R_2 = 15\,\mathrm{k\Omega}, \quad R_3 = R_4 = 10\,\mathrm{k\Omega}, \quad R_f = 30\,\mathrm{k\Omega}.$$

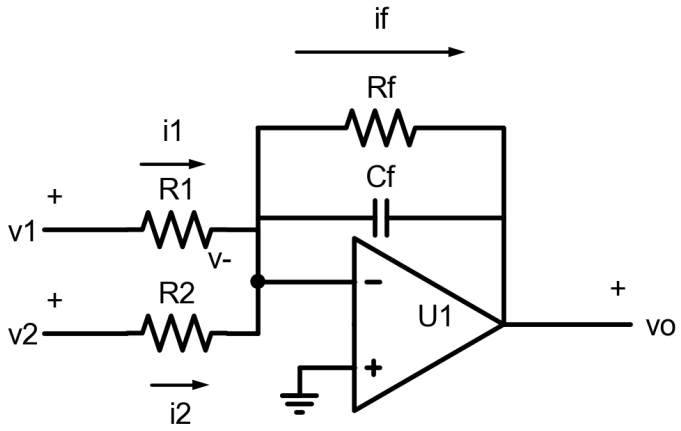
$$R_L = 1\,\mathrm{k\Omega}, \qquad V_{\mathrm{rail}} = \pm 10\,\mathrm{V}.$$

1. Assuming the op amp is not saturated, find v_+, and derive the relationship between v_o, v_1, v_2, and V_{ref}.
2. For $v_1 = 1\,\mathrm{V}$ and $v_2 = 2\,\mathrm{V}$, with R_L disconnected, find v_o, and find the op-amp output-terminal current.
3. For $v_1 = -1\,\mathrm{V}$ and $v_2 = 0\,\mathrm{V}$, with R_L disconnected, determine whether the output remains in the linear region. If not, state the saturation direction and the actual output voltage.

第 8 题

如下图所示理想运算放大器电路。C_f 待定，R_1、R_2 由设计要求确定。

$$R_f = 10\,\mathrm{k\Omega}.$$



1. 写出反馈阻抗 $Z_f(s)$，并写出 $V_o(s)$ 关于 $V_1(s)$、$V_2(s)$ 的表达式。
2. 按直流增益设计 R_1、R_2，使低频下

$$v_o \approx -2v_1 - \frac{1}{2}v_2.$$

3. 求由 $R_f \parallel C_f$ 引入的极点角频率，并说明高频时输出幅值的变化趋势。
4. 若希望截止角频率为

$$\omega_c = 1.0 \times 10^5 \,\mathrm{rad/s},$$

求 C_f，并给出对应的普通频率 f_c。

For the ideal operational-amplifier circuit shown in the figure, C_f is to be determined, and R_1, R_2 are set by the design requirements.

The value is

$$R_f = 10\,\mathrm{k\Omega}.$$

1. Write the feedback impedance $Z_f(s)$, and express $V_o(s)$ in terms of $V_1(s)$ and $V_2(s)$.
2. Choose R_1 and R_2 from the DC gains so that, at low frequency,

$$v_o \approx -2v_1 - \frac{1}{2}v_2.$$

3. Find the pole angular frequency introduced by $R_f \parallel C_f$, and describe how the output magnitude changes at high frequency.
4. If the desired cutoff angular frequency is

$$\omega_c = 1.0 \times 10^5 \,\mathrm{rad/s},$$

find C_f, and give the corresponding ordinary frequency f_c.

第 9 题

如下图所示理想运放电路，假设直流工作点已被复位且电容不存在漏电，分析由输入阶跃引起的小信号增量响应。除第 4 小问讨论限幅外，均按线性一阶运放模型计算。

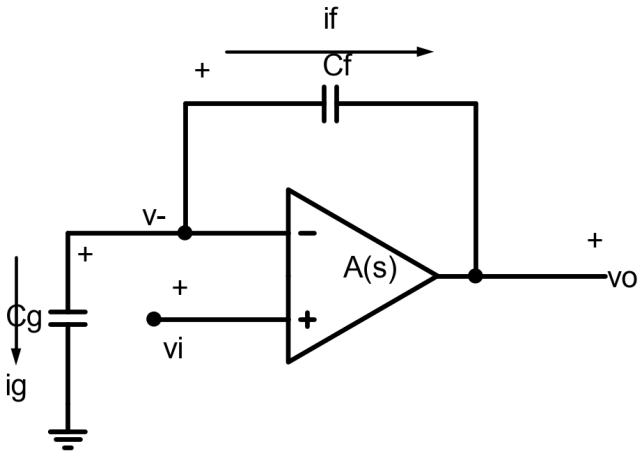
$$A(s) = \frac{A_0}{1 + s/\omega_p},$$

$$A_0 = 90, \qquad \omega_p = 100\,\text{rad/s},$$

$$C_g = 9\,\text{nF}, \qquad C_f = 1\,\text{nF}.$$

$$v_i(t) = u(t)\,\text{V}.$$

$$V_{\text{rail}} = \pm 12\,\text{V}.$$



1. 写出闭环传递函数 $H(s) = V_o(s)/V_i(s)$，并化为标准一阶形式。
2. 求闭环低频增量增益、闭环极点和时间常数。
3. 在输出不饱和的假设下，写出增量阶跃输入 $u(t)\,\text{V}$ 对应的 $v_o(t)$。
4. 判断第 3 小问的线性响应是否会碰到电源轨；若输入改为 $2u(t)\,\text{V}$，在理想电源轨限幅模型下写出输出响应，并求开始限幅的时间。

For the ideal op-amp circuit shown in the figure, assume the DC operating point has been reset and the capacitors have no leakage. Analyze the small-signal incremental response caused by the input step. Except for the rail-limiting discussion in part 4, use the linear first-order op-amp model.

The model and values are

$$A(s) = \frac{A_0}{1 + s/\omega_p},$$

$$A_0 = 90, \qquad \omega_p = 100\,\text{rad/s},$$

$$C_g = 9\,\text{nF}, \qquad C_f = 1\,\text{nF}.$$

$$v_i(t) = u(t)\,\text{V}.$$

$$V_{\text{rail}} = \pm 12\,\text{V}.$$

1. Derive the closed-loop transfer function $H(s) = V_o(s)/V_i(s)$, and express it in standard first-order form.
2. Find the closed-loop low-frequency incremental gain, closed-loop pole, and time constant.
3. Assuming the output does not saturate, write $v_o(t)$ for the incremental step input $u(t)\,\text{V}$.

4. Decide whether the linear response in part 3 reaches the supply rails. If the input is changed to $2u(t)\,\mathrm{V}$, write the output response under an ideal rail-limiting model and find the time at which rail limiting starts.

第 10 题

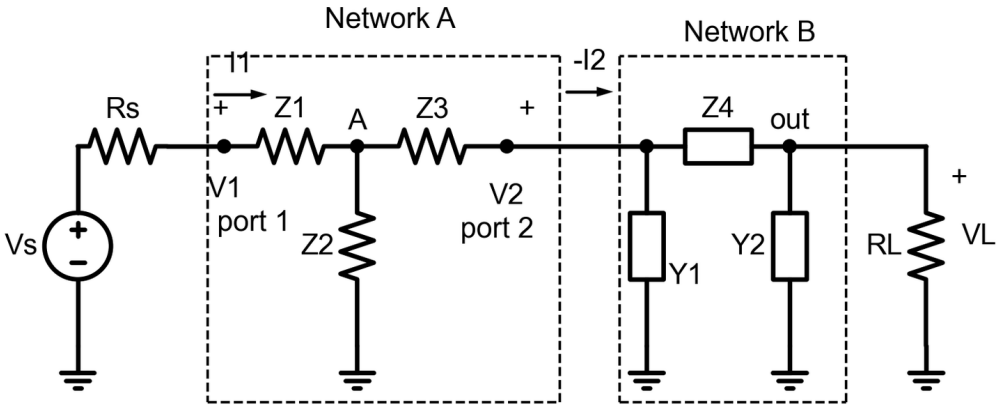
如下图所示，两个无源线性二端口从左到右级联。

传输参数定义为

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = T \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}.$$

元件与激励参数为

$$Z_1 = Z_2 = Z_3 = Z_4 = 1\,\mathrm{k\Omega}, \qquad Y_1 = Y_2 = 1\,\mathrm{mS},$$
$$V_s = 34\,\mathrm{V}, \qquad R_s = 1\,\mathrm{k\Omega}, \qquad R_L = 1\,\mathrm{k\Omega}.$$



1. 求电路 A 的 Z 参数矩阵，并将其转换为传输矩阵 T_A。
2. 用串联阻抗和并联导纳的基本传输矩阵求电路 B 的传输矩阵 T_B。
3. 求级联后的总传输矩阵 $T = T_A T_B$。
4. 在总二端口输出端接入 $R_L = 1\,\mathrm{k\Omega}$ 时，求从总二端口输入端看进去的输入阻抗。
5. 接入 V_s 与 R_s 后，求负载电压 V_L。

As shown in the figure, two passive linear two-port networks are cascaded from left to right.

The transmission parameters are defined by

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = T \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}.$$

The component and source values are

$$Z_1 = Z_2 = Z_3 = Z_4 = 1\,\mathrm{k\Omega}, \qquad Y_1 = Y_2 = 1\,\mathrm{mS},$$
$$V_s = 34\,\mathrm{V}, \qquad R_s = 1\,\mathrm{k\Omega}, \qquad R_L = 1\,\mathrm{k\Omega}.$$

1. Find the Z-parameter matrix of network A, and convert it to the transmission matrix T_A.

2. Use the elementary transmission matrices of series impedances and shunt admittances to find the transmission matrix T_B of network B.
3. Find the overall cascaded transmission matrix $T = T_A T_B$.
4. With $R_L = 1\,\mathrm{k}\Omega$ connected to the output port of the overall two-port, find the input impedance looking into the input port of the overall two-port.
5. After connecting V_s and R_s, find the load voltage V_L.